Translation confinement and angular propagation for a Reuleaux test family in Lebesgue's covering problem

Research draft
Author designation pending

9 October 2026

Abstract

For the hull of a diameter-one disk, a regular Reuleaux triangle and a rotated regular Reuleaux pentagon, we give explicit translation confinement using the area of the convex hull of two unequal disks. Every placement with area at most 35454253/42400000 has triangle translation norm less than 4/25 and pentagon translation norm less than 9/100. The true translation infimum is attained. Its clipping at this cutoff has angular Lipschitz constant 4270068/1750000, giving a conditional extension rule for certificates that already quantify over all translations. Applied to three inherited regional numerical premises, this rule extends the certified closed angle set at threshold 1669/2000, but leaves two positive gaps. We also describe the precise scope of retained restricted cell certificates. No new global numerical lower bound or priority claim is made.

1 Questions, test family and main results

A convex universal cover is a compact convex planar set into which every planar set of diameter at most one can be moved by a rigid motion. Write a_{Leb} for the infimum of the areas of such covers. The following finite test family yields a necessary condition for being a universal cover. We address three auxiliary questions: explicit translation confinement for low-area placements, angular stability of a clipped placement infimum, and propagation of a regional lower bound to nearby angles.

Definition 1.1. Let $\mathbb{D}(p,r) = \{x \in \mathbb{R}^2 : \|x-p\| \le r\}$, and put

$$D = \mathbb{D}(0,1/2), \qquad \rho_n = \frac{1}{2\cos(\pi/(2n))},$$
$$c_{n,j} = \rho_n(\cos(2\pi j/n), \sin(2\pi j/n)), \qquad B_n = \bigcap_{j=0}^{n-1} \mathbb{D}(c_{n,j},1), \quad n = 3,5. \tag{1}$$

Let R_ϕ be rotation through angle ϕ about the origin. For independent $t_3 = (x_3,y_3)$ and $t_5 = (x_5,y_5)$ in $\mathbb{R}^2$, define

$$\begin{aligned} K(t_3,t_5,\phi) &= \operatorname{conv}\big(D \cup (t_3+B_3) \cup (t_5+R_\phi B_5)\big),\\ H(t_3,t_5,\phi) &= \operatorname{area} K(t_3,t_5,\phi),\\ G(\phi) &= \inf_{(t_3,t_5)\in\mathbb{R}^2\times\mathbb{R}^2} H(t_3,t_5,\phi), \qquad M = \inf_{\phi\in\mathbb{R}} G(\phi). \end{aligned} \tag{2}$$

All angle measures are in radians. The target threshold in the regional application is $\tau = 1669/2000 = 0.8345$.

The bodies B_3 and B_5 are the regular Reuleaux bodies of constant width one, hence diameter one. Gibbs explored lower bounds using constant-width test bodies [1, Section 2]; Mishra uses the present three-body family [2, Section 1.1]. To see $a_{\mathrm{Leb}} \geq M$, place the three test bodies inside any convex universal cover. A common motion centers its placed disk and fixes the triangle's orientation; the pentagon then has a relative angle ϕ and the two bodies have independent translations. Reflections cause no extra shape parameter, since each centered body in (1) is invariant under horizontal reflection. Convexity of the cover contains their hull. Thus its area is at least H for that placement and hence at least M.

Lemma 1.2 (Angle reduction)**.** *The function G is even and has period $2\pi/15$. In particular, $M = \inf_{\phi\in[0,\pi/15]} G(\phi)$.*

Proof. Rotation by $2\pi/n$ permutes the defining centers of B_n. The pentagon therefore gives period $2\pi/5$ for G. A common rotation by $2\pi/3$ preserves D and B_3, and maps each independent translation bijectively; it gives the other period $2\pi/3$. The integer combination $2(2\pi/3) - 3(2\pi/5) = 2\pi/15$ is a period. Horizontal reflection S preserves D, B_3, B_5 and satisfies $SR_\phi = R_{-\phi}S$, so it gives $G(-\phi) = G(\phi)$ after the bijection $(t_3, t_5) \mapsto (St_3, St_5)$. Periodicity and evenness give the claimed reduction. These are symmetries of the infimum; they assert no symmetry of an individual minimizer. □

Relation to prior work. Gibbs's numerical exploration of relative placements motivates the family; sampled placement areas give upper bounds on its minimum and must not be read as certified lower bounds [1, Section 2]. Mishra's version 2 proves the all-placement lower bound 0.8344 for the same family using confinement, erosion and finite-box certification [2, Sections 4.2, 5.1 and 5.5]. The analytic radii printed in that paper's Remark 20 are approximately 0.192366 and 0.195891 below 0.8344. The radii 0.16 and 0.09 below are smaller and apply up to the higher cutoff $L_3 \approx 0.8361852$. This comparison concerns those printed analytic estimates; it establishes neither priority nor a computational saving.

Zeng's version 1 proves a global 0.834 bound and gives radial collars beginning at 0.08 at that threshold, with a remaining square core [3, Section S11]. Their stated threshold alone does not transfer those exclusions to 0.8345 or to L_3. Zeng also supplies the disk-plus-point coercivity and Hausdorff-area transfer tools used here [3, Lemma S5.6 and Section S6.4]. A comparison with the exact regional data in these papers remains open.

Our purpose is to record explicit rational estimates and their conditional regional consequences using classical convex-geometric tools. We do not claim a new global covering bound or priority for these specializations. The partial support certificates in the supplement do not resolve Mishra's global dual or variational question [2, Section 7].

The main analytic result is as follows; its proof occupies the next two sections.

Theorem 1.3. *Set*

$$d_3 = \frac{4}{25}, \qquad d_5 = \frac{9}{100}, \qquad L_3 = \frac{35454253}{42400000}, \qquad L_5 = \frac{1995480341}{2385000000},$$
$$Q = \mathbb{D}(0, d_3) \times \mathbb{D}(0, d_5), \qquad C = \frac{4270068}{1750000}. \tag{3}$$

For every real angle and every independent translation pair,

$$\begin{aligned} \|t_3\| \geq d_3 &\implies H(t_3, t_5, \phi) > L_3 > \tau, \\ \|t_5\| \geq d_5 &\implies H(t_3, t_5, \phi) > L_5 > L_3. \end{aligned} \tag{4}$$

The infimum defining $G(\phi)$ is attained for every ϕ, and M is attained at a translation pair and an angle in $[0, \pi/15]$. If $F(\phi) = \min_{t \in Q} H(t, \phi)$ and $b(\phi) = \min\{L_3, G(\phi)\}$, then

$$b(\phi) = \min\{L_3, F(\phi)\}, \qquad |b(\phi) - b(\psi)| \le C|\phi - \psi|. \tag{5}$$

Every placement with $H \le L_3$ lies in the strict interior of Q.

2 Two unequal disks and rational confinement

Lemma 2.1 (Disk sandwich)**.** *For $n = 3, 5$,*

$$\mathbb{D}(0, 1 - \rho_n) \subset B_n \subset \mathbb{D}(0, \rho_n). \tag{6}$$

Moreover $\rho_3 < 289/500$ and $\rho_5 < 263/500$. Consequently the inner disks of radii $r_3 = 211/500$ and $r_5 = 237/500$ lie in B_3 and B_5, respectively.

Proof. If $\|x\| \le 1 - \rho_n$, then $\|x - c_{n,j}\| \le \|x\| + \rho_n \le 1$ for every j. For the outer inclusion, write $x = su$ with $\|u\| = 1$ and $s > \rho_n$. Some defining center has angular distance at most π/n from $-u$, so

$$\|x - c_{n,j}\|^2 \ge s^2 + \rho_n^2 + 2s\rho_n \cos(\pi/n) > 2\rho_n^2(1 + \cos(\pi/n)) = 1.$$

Here $\cos(\pi/n) > 0$, and the last identity follows from the definition of ρ_n. Thus $x \notin B_n$.

For the triangle $\rho_3 = 1/\sqrt{3}$ and $500^2 = 250000 < 3 \cdot 289^2 = 250563$. For the pentagon, $\cos^2(\pi/10) = (5 + \sqrt{5})/8$. Since $223^2 = 49729 < 50000$, we have $\sqrt{5} > 223/100$ and $\cos^2(\pi/10) > 723/800$. Finally $723 \cdot 263^2 - 800 \cdot 250^2 = 9187 > 0$, so $\cos(\pi/10) > 250/263$ and $\rho_5 < 263/500$. The inner radii follow from (6). Centered disk inclusions persist under rotation. □

Lemma 2.2 (Unequal-disk hull)**.** *Let $R \ge r > 0$ and let $A(R, r, d) = \operatorname{area} \operatorname{conv}(\mathbb{D}(0, R) \cup \mathbb{D}((d, 0), r))$. For $d > R - r$,*

$$A(R, r, d) = \frac{\pi}{2}(R^2 + r^2) + (R^2 - r^2) \arcsin \frac{R - r}{d} + (R + r)\sqrt{d^2 - (R - r)^2}. \tag{7}$$

For fixed R, r, A is nondecreasing on $d \ge 0$; for $d \le R - r$ it is πR^2.

Proof. Put $q = (R - r)/d$ and $\theta = \arccos q$. The common external tangent normals are $(q, \pm \sin\theta)$, with contact points

$$P_\pm = R(q, \pm \sin\theta), \qquad Q_\pm = (d, 0) + r(q, \pm \sin\theta).$$

Figure 1 shows the resulting boundary. Traversing it counterclockwise gives the large-circle arc from θ to $2\pi - \theta$, the bottom segment P_-Q_-, the small-circle arc from $-\theta$ to θ, and the top segment Q_+P_+. In $\frac{1}{2}\int(x\,dy - y\,dx)$ their respective contributions are

$$R^2(\pi - \theta), \quad \frac{dR\sin\theta}{2}, \quad r^2\theta + dr\sin\theta, \quad \frac{dR\sin\theta}{2}.$$

Their sum is $\pi R^2 - (R^2 - r^2)\theta + d(R + r)\sin\theta$. Substitute $\theta = \pi/2 - \arcsin q$ and $d\sin\theta = \sqrt{d^2 - (R - r)^2}$. This description applies both when the disks overlap and when they are disjoint.

For monotonicity, if $0 \le d_1 \le d_2$ and $d_2 > 0$, put $\lambda = d_1/d_2$. Then

$$\mathbb{D}((d_1, 0), r) = (1 - \lambda)\mathbb{D}(0, r) + \lambda\mathbb{D}((d_2, 0), r) \subset \operatorname{conv}(\mathbb{D}(0, R) \cup \mathbb{D}((d_2, 0), r)).$$

The hull at d_1 is contained in the hull at d_2. The case $d_2 = 0$ is immediate, and the containment formula for $d \le R - r$ is elementary. □

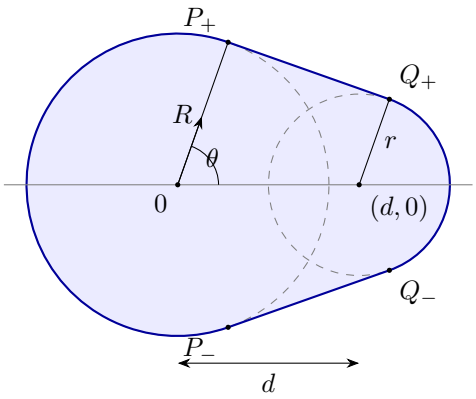


Figure 1: The boundary used in the unequal-disk area formula. The drawing uses the illustrative values $R = 1$, $r = 0.6$, $d = 1.2$; the contact points and tangents are drawn at that scale. Dashed arcs lie inside the hull.

We use the classical rational enclosure

$$\frac{333}{106} < \pi < \frac{22}{7}. \tag{8}$$

Lucas gives positive-integral identities for both inequalities [4, Sections 1 and 3 of the cited preprint].

Proof of the confinement assertions in Theorem 1.3. The hull contains D and the translated inner disk from Lemma 2.1. After rotation of that pair,

$$H(t_3, t_5, \phi) \geq A(1/2, r_n, \|t_n\|) \geq A(1/2, r_n, d_n) \quad \text{when } \|t_n\| \geq d_n.$$

Use $\arcsin z \geq z$ for $0 \leq z < 1$, which follows by integrating $(1-z^2)^{-1/2} \geq 1$, and the enclosure (8). For $n = 3$, $1/2 - r_3 = 39/500$, $d_3 = 80/500$, and

$$\sqrt{d_3^2 - (1/2 - r_3)^2} = \frac{\sqrt{4879}}{500} > \frac{349}{2500}, \qquad 25 \cdot 4879 = 121975 > 349^2 = 121801.$$

Formula (7) now gives the strict rational estimate

$$\begin{aligned} A(1/2, r_3, d_3) &> \frac{107021 \cdot 333}{53000000} + \frac{17979}{250000}\frac{39}{80} + \frac{461}{500}\frac{349}{2500} \\ &= \frac{35637993}{53000000} + \frac{701181}{20000000} + \frac{160889}{1250000} = L_3. \end{aligned}$$

For $n = 5$, $1/2 - r_5 = 13/500$, $d_5 = 45/500$, and $\sqrt{1856}/500 > 43/500$ because $1856 > 43^2 = 1849$. Consequently

$$\begin{aligned} A(1/2, r_5, d_5) &> \frac{118669 \cdot 333}{53000000} + \frac{6331}{250000}\frac{13}{45} + \frac{487}{500}\frac{43}{500} \\ &= \frac{39516777}{53000000} + \frac{82303}{11250000} + \frac{20941}{250000} = L_5. \end{aligned}$$

Finally,

$$L_3 - \tau = \frac{71453}{42400000} > 0, \quad L_5 - \tau = \frac{5197841}{2385000000} > 0, \quad L_5 - L_3 = \frac{4714439}{9540000000} > 0.$$

These strict estimates include the norm cutoff faces themselves. Their contrapositives give the strict interior assertion for $H \leq L_3$. □

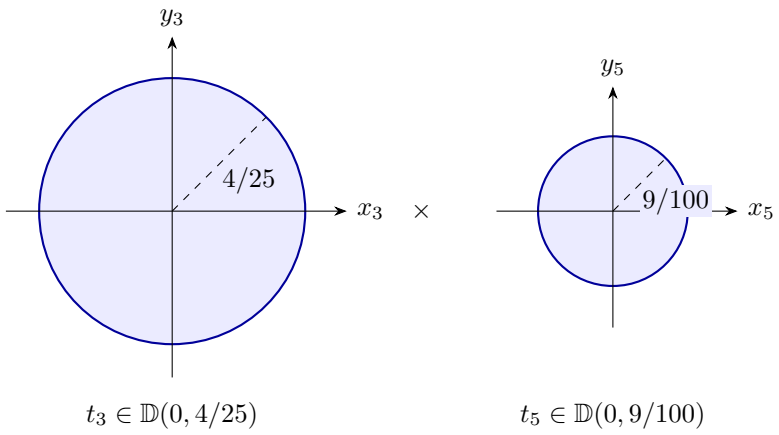


Figure 2: The compact product Q uses independent translation planes; the two radii are shown on a common linear scale. Every placement with $H \le L_3$ lies strictly inside both disks. Their vertical-coordinate projection is the rectangle V of Corollary 2.3.

Corollary 2.3 (Vertical or radial reduction)**.** *Let I be any angle set and $\kappa \le L_3$. A bound $H((x_3, y_3), (x_5, y_5), \phi) \ge \kappa$ for all $x_3, x_5 \in \mathbb{R}$, all*

$$(y_3, y_5) \in V = [-4/25, 4/25] \times [-9/100, 9/100], \quad \phi \in I$$

implies the same bound for all four translation coordinates and $\phi \in I$. Alternatively, a bound on all of Q transfers to all translations at the same threshold.

Proof. Outside the closed rectangle V, one vertical coordinate exceeds its norm cutoff in absolute value, and (4) applies. Outside Q the radial version applies directly. □

The rectangle's area relative to the historical vertical square $[-7/10, 7/10]^2$ is

$$\frac{(8/25)(9/50)}{(7/5)^2} = \frac{36}{1225}.$$

This is a domain ratio, with no assertion about computational success on that domain.

3 Attainment and a modulus for the clipped true infimum

For a compact convex planar body K with nonempty interior, write $w_K(\theta)$ for its width in direction $(\cos\theta, \sin\theta)$. The following standard formulas control the area even when the hull's boundary type changes.

Lemma 3.1 (Classical area formulas and their application)**.** *For $\varepsilon \ge 0$,*

$$\operatorname{area}(K + \varepsilon\mathbb{D}(0,1)) = \operatorname{area} K + \operatorname{Per}(K)\varepsilon + \pi\varepsilon^2, \qquad \operatorname{Per}(K) = \int_0^\pi w_K(\theta)\, d\theta. \tag{9}$$

In particular, $K \subset \mathbb{D}(0, R)$ implies $\operatorname{Per}(K) \le 2\pi R$. If K, L have Hausdorff distance at most ε, then

$$|\operatorname{area} K - \operatorname{area} L| \le \max\{\operatorname{Per}(K), \operatorname{Per}(L)\}\varepsilon + \pi\varepsilon^2. \tag{10}$$

Proof. The first identity is the planar Steiner formula [5, Theorem 8.6.1 and Corollary 12.3.2]; the second is Cauchy's projection formula in dimension two [6, Theorem 1]. In the latter, one-dimensional projection length is width, and opposite directions give the same width. Thus $w_K \le 2R$ gives the perimeter bound. Hausdorff distance at most ε gives $K \subset L + \varepsilon\mathbb{D}(0,1)$ and the reverse inclusion with K, L interchanged. Apply Steiner's formula in both directions and use monotonicity of area to obtain (10). □

Lemma 3.2 (Continuity and compact minimization)**.** *The function H is continuous, and the attainment assertions of Theorem 1.3 hold.*

Proof. Rigid motions of the fixed compact bodies vary continuously in Hausdorff distance. If corresponding constituent sets are within ε, their convex hulls are also within ε: replace the points in any finite convex combination by corresponding nearby points. Locally the hulls lie in a fixed disk and contain D. Their perimeters are uniformly bounded by Lemma 3.1, so (10) proves continuity of H.

Both B_n contain the origin by Lemma 2.1, so the hull contains D and each center t_n. The disk-plus-point estimate of Zeng [3, Lemma S5.6] therefore applies: the triangle with vertex t_n and the perpendicular diameter of D has area $\|t_n\|/2$. Consequently

$$H(t_3, t_5, \phi) \ge \frac{1}{2}\max\{\|t_3\|, \|t_5\|\}. \tag{11}$$

At zero translations the hull lies in $\mathbb{D}(0, \rho_3)$, since $\rho_5 < \rho_3$ and $\rho_3 > 1/2$. Thus $H(0,0,\phi) \le \pi/3 < 3/2$ for every angle. Outside $T = \mathbb{D}(0,3) \times \mathbb{D}(0,3)$, (11) gives $H > 3/2$. Consequently the translation infimum for each angle equals a minimum over the compact set T. Continuity proves attainment. The same argument on $T \times [0, \pi/15]$ proves attainment of the joint minimum, which equals M by Lemma 1.2. If a minimizing placement has value at most L_3, confinement puts it in the strict interior of Q. □

Proof of the clipped modulus in Theorem 1.3. Every placement outside Q has area greater than L_3. Its infimum there is therefore at least L_3, even if equality is approached as a limit. Splitting the unrestricted infimum over Q and its complement gives

$$\min\{L_3, G(\phi)\} = \min\{L_3, F(\phi)\}. \tag{12}$$

Compactness of Q and continuity make F a minimum. We do not assert $G = F$ when their values are above the clipping level.

Fix $t \in Q$. All constituent points, and hence their hull, lie in $\mathbb{D}(0, R_*)$ at every angle, where

$$R_* = \frac{369}{500}, \qquad d_3 + \rho_3 < \frac{80+289}{500}, \qquad d_5 + \rho_5 < \frac{45+263}{500} < R_*.$$

Let $c_* = 263/500$. Rotating a point of B_5 through δ moves it by at most $\rho_5|\delta| \le c_*|\delta|$; the translation t_5 is fixed. Thus the two hulls have Hausdorff distance at most $c_*|\delta|$, and each has perimeter at most $2\pi R_*$. Divide an arbitrary angle difference $\delta = \psi - \phi$ into m equal steps. Apply (10) at each step and add to obtain

$$|H(t,\phi) - H(t,\psi)| \le 2\pi R_* c_* |\delta| + \frac{\pi c_*^2 \delta^2}{m}.$$

All intermediate hulls have the same radius bound. Letting $m \to \infty$ proves a uniform Lipschitz bound on Q. The enclosure (8) gives

$$2\pi R_* c_* < \frac{44}{7}\frac{369}{500}\frac{263}{500} = \frac{4270068}{1750000} = C.$$

i	closed interval I_i	display of L_i	display of η_i
0	$[0, 3/1250]$	0.8345012284208383	$5.0344314587 \times 10^{-7}$
10	$[99/10000, 101/10000]$	0.8345054097006209	$2.2170551116 \times 10^{-6}$
end	$[\pi/15 - 1/125, \pi/15]$	0.8345962664387585	$3.9452830218 \times 10^{-5}$

Table 1: The three closed source intervals and approximate displays. The exact values of L_i and the halo radii $\eta_i = (L_i - \tau)/C$ are in the generated data and the mathematical supplement.

A common Lipschitz bound passes to a minimum: evaluating at a minimizer for either angle gives both one-sided inequalities for F. The map $s \mapsto \min\{L_3, s\}$ is 1-Lipschitz. Together with (12), this proves (5). □

Corollary 3.3 (Conditional angular propagation)**.** *Suppose I is a nonempty closed interval and an all-translation certificate gives $G(\psi) \geq L$ for every $\psi \in I$. Then for every $\phi \in \mathbb{R}$,*

$$G(\phi) \geq \min\{L, L_3\} - C \operatorname{dist}(\phi, I). \tag{13}$$

If $\min\{L, L_3\} > \tau$, the certificate extends to the closed distance neighborhood of radius $(\min\{L, L_3\} - \tau)/C$ at threshold τ. A finite collection of such neighborhoods covering $[0, \pi/15]$ would prove $M \geq \tau$ and $a_{\mathrm{Leb}} \geq \tau$.

Proof. On I, $b \geq \min\{L, L_3\}$. Apply (5) between ϕ and each $\psi \in I$, take the best resulting lower bound, and use $G \geq b$. The final implication uses Lemma 1.2. □

Restricted translation certificates do not supply the hypothesis of this corollary. No finite covering of the full reduced angle interval has been established here.

4 Conditional application to three inherited regional premises

Assume the three inherited regional premises

$$(\mathcal{P}_i): \quad H(t_3, t_5, \phi) \geq L_i \quad \text{for every } t_3, t_5 \in \mathbb{R}^2 \text{ and every } \phi \in I_i, \quad i = 0, 10, \mathrm{end}. \tag{14}$$

Their exact rational constants are supplied in the mathematical supplement and in the computational companion described below. The rounded displays in Table 1 are reading aids. These are inherited numerical-geometry premises. The companion supplies finite witnesses and checking programs for their numerical ingredients. The present application remains conditional on the three stated premises; the supplement explains the geometric implications of the finite checks.

The inherited exact data satisfy $\tau < L_i < L_3$. Define

$$\begin{aligned} \eta_i &= \frac{L_i - \tau}{C}, \quad i = 0, 10, \mathrm{end}, \\ a_0 &= \frac{3}{1250} + \eta_0, & a_{10} &= \frac{99}{10000} - \eta_{10}, \\ b_{10} &= \frac{101}{10000} + \eta_{10}, & a_{\mathrm{end}} &= \frac{\pi}{15} - \frac{1}{125} - \eta_{\mathrm{end}}. \end{aligned} \tag{15}$$

Proposition 4.1 (Regional implication)**.** *Under (14), every independent translation pair satisfies $H \geq \tau$ on the closed set*

$$E = [0, a_0] \cup [a_{10}, b_{10}] \cup [a_{\mathrm{end}}, \pi/15]. \tag{16}$$

The same placements satisfy $H > \tau$ on

$$E_{\mathrm{strict}} = [0, a_0) \cup (a_{10}, b_{10}) \cup (a_{\mathrm{end}}, \pi/15]. \tag{17}$$

The complement of this closed threshold union in $[0, \pi/15]$ is $(a_0, a_{10}) \cup (b_{10}, a_{\mathrm{end}})$, with positive widths

$$w_1 = \frac{3}{400} - \eta_0 - \eta_{10}, \qquad w_2 = \frac{\pi}{15} - \frac{181}{10000} - \eta_{10} - \eta_{\mathrm{end}}. \tag{18}$$

The additional and total covered lengths are, respectively,

$$\eta_0 + 2\eta_{10} + \eta_{\mathrm{end}}, \qquad \frac{53}{5000} + \eta_0 + 2\eta_{10} + \eta_{\mathrm{end}}. \tag{19}$$

Proof. Corollary 3.3 gives the envelopes $e_i(\phi) = L_i - C\,\mathrm{dist}(\phi, I_i)$, since $L_i < L_3$. The condition $e_i \geq \tau$ is exactly $\mathrm{dist}(\phi, I_i) \leq \eta_i$; strict inequality holds at distances less than η_i. Intersect these neighborhoods with the reduced interval to get (16) and (17). The fundamental endpoints $0, \pi/15$ lie in the original intervals, whose bounds are strictly above τ. Finally $H \geq G$ transfers the conclusion to every translation.

The exact retained radius brackets are

$$\frac{100688629}{200000000000000} < \eta_0 < \frac{251721573}{500000000000000}$$
$$\frac{2217055111}{1000000000000000} < \eta_{10} < \frac{277131889}{125000000000000}$$
$$\frac{19726415109}{500000000000000} < \eta_{\mathrm{end}} < \frac{39452830219}{1000000000000000}$$

In particular $\eta_0 < 10^{-6}$, $\eta_{10} < 10^{-5}$ and $\eta_{\mathrm{end}} < 10^{-4}$. Thus

$$w_1 > \frac{3}{400} - \frac{11}{10^6} > \frac{7}{1000} > 0, \qquad w_2 > \frac{1}{5} - \frac{181}{10000} - \frac{11}{100000} = \frac{18179}{100000} > \frac{9}{50} > 0,$$

where $\pi > 3$ follows from the enclosure (8). Subtracting the adjacent endpoints gives (18). The first and last sources acquire one inward halo after intersection with the reduced interval, and the middle source acquires two. Positivity of both gaps makes the three closed intervals disjoint. Their lengths therefore add as in (19). □

The recorded rational-C envelopes equal τ at the four new outer faces $a_0, a_{10}, b_{10}, a_{\mathrm{end}}$. We retain the conservative $\geq \tau$ statement there; this envelope does not infer strictness at those faces. This is no assertion of equality for the actual G or H. Indeed the proof of the modulus also supplies the smaller real constant $2\pi R_* c_* < C$. The complement of the stated strict guarantee is $[a_0, a_{10}] \cup [b_{10}, a_{\mathrm{end}}]$, whereas the complement of E consists of open gaps.

The covered length is about 0.01064439038 radians, approximately 5.08% of $[0, \pi/15]$. The halos add about 0.00004439038 radians. The gap widths are about 0.00749727950 and 0.19129784035 radians. These displays describe this particular union; they do not say that other certificates cannot apply in the gaps, or that the threshold fails there. In particular, this application gives no global bound $M \geq \tau$.

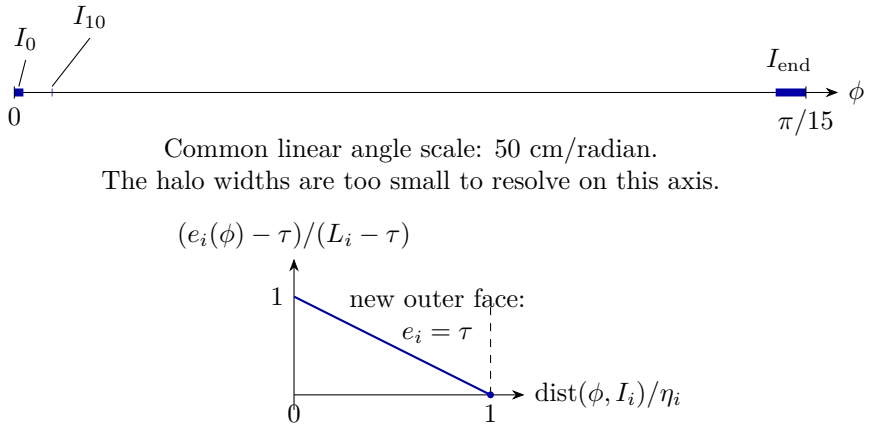


Figure 3: Conditional source coverage and a normalized halo envelope, assuming the three inherited premises (14). The top source intervals use a common linear scale. The inset rescales each source's margin and radius separately, showing why the new face is included in the conservative closed threshold guarantee.

5 What remains unresolved

The confinement theorem answers a uniform exclusion question for low-area placements. The clipped modulus answers an angular stability question, and the propagation corollary gives an explicit consequence of any valid all-translation regional lower bound. Their proofs do not require the three inherited numerical premises. The numerical application does: it covers about 5.08% of the reduced angle interval at $\tau = 0.8345$ and leaves two positive gaps. These gaps describe missing coverage, not counterexamples to the threshold.

The mathematical supplement retains 33 restricted A cells and two restricted B cells, with unrestricted horizontal translations but restricted vertical and angular domains. The A cells leave 24 residual closed leaves in the required vertical rectangle. None of these restricted certificates supplies an all-translation premise for Corollary 3.3. Their support mechanism is a specialization of common-direction shadow-system convexity [7, 8], supporting planes and nonnegative exact weights.

The supplement also records an all-translation endpoint pilot at $L_B \approx 0.8341743532983762$. This bound is already implied numerically by Mishra's stronger all-placement 0.8344 result [2]; its role here is to illustrate an inner-polygon symmetrization argument. It adds no threshold-τ coverage. Neither this example nor the restricted cells complete a global dual certificate.

The comparison made here is bounded to the cited versions and stated domains. Exact regional non-subsumption and priority for the explicit specializations have not been established. The resulting note provides explicit estimates and conditional partial certification, without a claim to have solved a named published open problem or improved the established global covering bound.

Supplement and computational companion. The supplement contains the restricted certificate mechanism, exact constants and cell tables. Finite checking programs, their mathematical witness data and run instructions are available in the Universal covering computational companion (version `companions-v2`). The checks cover the common-core and middle-strip certificates, the restricted cells, the endpoint pilot and the exact angular-propagation arithmetic. The supplement specifies their mathematical scope. The accompanying Lean development checks its encoded implications

and rational identities under explicit remaining hypotheses; it does not certify the companion's geometric computations. The work was developed with AI assistance. Human author designation remains pending.

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